

claude-windows / 42211f80-2169-4025-ad9d-99c1aa29ff70

Metadata

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- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\42211f80-2169-4025-ad9d-99c1aa29ff70\subagents\workflows\wf_6237b110-d58\agent-a29c0209e9090f354.jsonl`  
- SHA-256: `cef99a78a33c61f7f6a113cb468196bee6331c79c5ff73a17df9a93547f8e8d2`  
- Source modified: `2026-06-06T17:34:14+00:00`  
- Imported at: `2026-07-05T16:48:28+00:00`  
- project: `wf_6237b110-d58`  
- session_id: `42211f80-2169-4025-ad9d-99c1aa29ff70`
```

Transcript

1. user (2026-06-06T17:33:44.905Z)

Adversarial Claim Verifier (voter 1/3)

Be SKEPTICAL. Try to REFUTE this claim. ?2/3 refutations kill it.

Research question

Research alternative methods to automatically detect rally start/end segments (temporal segmentation) in racket-sport video ? especially badminton ? for a SELF-HOSTED, LOCAL-FIRST pipeline, and assess whether a cloud-VLM teacher ? local-model distillation is the best approach or whether other methods are competitive or complementary. Keep it focused ("mini" research): prioritize actionable, commercially-licensable, locally-runnable approaches over exotic SOTA needing huge compute or restrictive licenses.

OUR CONTEXT / CONSTRAINTS (ground every recommendation in these):

- Self-hosted on a single consumer GPU (RTX 2070 Super), Windows + WSL2. We want LOCAL-ONLY inference at runtime ? privacy lever: user video never leaves the device. It's a commercial product, so PERMISSIVE / commercial-friendly licenses are REQUIRED (avoid AGPL and non-commercial-only weights/datasets).
- Current substrate: WASB (MIT) shuttle-trajectory detector ? per-frame (x, y, visibility). We derive rally windows via velocity/merge-gap/min-duration thresholds plus a net-crossing gate. We ALSO have a permissive person detector available (torchvision Faster R-CNN + supervision ByteTrack) and the video's AUDIO track.
- Current best approach: use paid Gemini (cloud VLM, full-context) OFFLINE as a TEACHER to label rallies, then DISTILL into a local model (logistic regression on shuttle-trajectory features). Findings so far: threshold-tuning ceilings around F1 ? 0.44 and overfits with few videos; the learned model is data-starved at n=2 labeled videos; WASB candidate RECALL is a real bottleneck (its windows sometimes don't align with true rallies at IoU 0.5).
- Inputs we can exploit locally: video frames, the shuttle trajectory, player bounding boxes / pose (if we add a permissive pose model), and the AUDIO track (shuttle "thwack" hits, crowd noise, umpire/score calls).

SPECIFICALLY COVER:

1. Method families for rally/point segmentation in racket sports (badminton, tennis, table tennis, pickleball): audio-based shot/hit detection; temporal action localization (TAL) deep models; pose/player-stance + court-homography state machines (e.g., "serve-ready stance" gating); classical change-point detection / HMM on the shuttle trajectory; scoreboard / score-overlay OCR for point boundaries; self-supervised / weakly-supervised approaches; and VLM teacher?student distillation.
2. For EACH family: how it works, the labels/data it needs, expected accuracy, local-inference COMPUTE cost, and LICENSE / commercial viability of any models, datasets, or weights involved.
3. Badminton rally-segmentation datasets (e.g., ShuttleSet, TrackNet/TrackNetV2 data) and their licenses / commercial usability.
4. A RANKED recommendation for our local-first, permissive, single-GPU setup: is Gemini-teacher ? local-distillation the best, or are some alternatives competitive or complementary ?

especially AUDIO-based hit detection and pose/court-geometry state machines, which we can run fully locally?

5. Concrete, cheap next experiments we could run with what we already have (shuttle trajectory + person detector + audio).

Claim under review

"ShuttleSet is a human-annotated stroke-level badminton singles dataset containing 104 sets, 3,685 rallies, and 36,492 strokes across 44 matches (2018-2021), with stroke-level annotations including 18 shot-type classes, hitting locations, and both players' locations at each stroke."

****Source:**** <https://arxiv.org/abs/2306.04948> (primary)

****Supporting quote:**** "104 sets, 3,685 rallies, and 36,492 strokes in 44 matches between 2018 and 2021"

Checklist

1. Is the claim actually supported by the quote, or is it an overreach/misread?
2. WebSearch for contradicting evidence ? does any credible source dispute or heavily qualify this?
3. Is the source quality sufficient for the claim's strength? (extraordinary claims need primary sources)
4. Is the claim outdated? (check dates ? old claims about fast-moving fields are suspect)
5. Is this a marketing claim / press release / cherry-picked benchmark / forum speculation?

****refuted=true**** if: unsupported by quote / contradicted / low-quality source for strong claim / outdated / marketing fluff.

****refuted=false**** ONLY if: claim is well-supported, current, and source quality matches claim strength.

Default to refuted=true if uncertain.

Structured output only. Evidence MUST be specific.

2. assistant (2026-06-06T17:33:47.477Z)

I'll verify this claim about ShuttleSet. Let me check the primary source and search for contradicting evidence.

3. assistant (2026-06-06T17:33:51.631Z)

Let me verify the specific numbers in the claim against the primary source.

4. user (2026-06-06T17:33:57.379Z)

ShuttleSet Dataset Statistics

****Dataset Composition:****

- 104 sets
- 3,685 rallies
- 36,492 strokes
- 44 matches between 2018 and 2021
- 27 top-ranking men's and women's singles players

****Shot Classification:****

- 18 distinct shot type classes

****Stroke-Level Annotations:****

According to the abstract, the dataset includes annotations for:

- "shot type with a choice of 18 distinct classes, the corresponding hitting locations, and the locations of both players at each stroke"

The annotations were created using "a computer-aided labeling tool" to enhance efficiency and effectiveness in selecting and recording stroke information.

5. user (2026-06-06T17:33:59.290Z)

Web search results for query: "ShuttleSet badminton dataset 104 sets 3685 rallies 36492 strokes 44 matches"

Links: [{"title": "ShuttleSet: A Human-Annotated Stroke-Level Singles Dataset for Badminton Tactical Analysis | DeepAI", "url": "https://deepai.org/publication/shuttle-set-a-human-annotated-stroke-level-singles-dataset-for-badminton-tactical-analysis"}, {"title": "ShuttleSet: A Human-Annotated Stroke-Level Singles Dataset for Badminton Tactical Analysis - ADS", "url": "https://ui.adsabs.harvard.edu/abs/2023arXiv230604948W/abstract"}, {"title": "[2306.04948] ShuttleSet: A Human-Annotated Stroke-Level Singles Dataset for Badminton Tactical Analysis", "url": "https://arxiv.org/html/2306.04948"}, {"title": "[2306.04948] ShuttleSet: A Human-Annotated Stroke-Level Singles Dataset for Badminton Tactical Analysis", "url": "https://arxiv.org/abs/2306.04948"}, {"title": "Google Scholar", "url": "https://scholar.google.com/scholar?q=ShuttleSet:+A+Human-Annotated+Stroke-Level+Singles+Dataset+for+Badminton+Tactical+Analysis."}, {"title": "(PDF) ShuttleSet: A Human-Annotated Stroke-Level Singles Dataset for Badminton Tactical Analysis", "url": "https://www.researchgate.net/publication/371413847_ShuttleSet_A_Human-Annotated_Stroke-Level_Singles_Dataset_for_Badminton_Tactical_Analysis"}, {"title": "ShuttleSet: A Human-Annotated Stroke-Level Singles Dataset for Badminton Tactical Analysis - National Yang Ming Chiao Tung University Academic Hub", "url": "https://scholar.nycu.edu.tw/en/publications/shuttle-set-a-human-annotated-stroke-level-singles-dataset-for-bad/"}, {"title": "ShuttleSet: A Human-Annotated Stroke-Level Singles Dataset for Badminton Tactical Analysis | Proceedings of the 29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining", "url": "https://dl.acm.org/doi/10.1145/3580305.3599906"}, {"title": "ShuttleSet: A Human-Annotated Stroke-Level Singles Dataset for Badminton Tactical Analysis", "url": "https://arxiv.org/pdf/2306.04948"}, {"title": "S 2 -Labeling: Shot-By-Shot Microscopic Badminton Singles Tactical Dataset | Request PDF", "url": "https://www.researchgate.net/publication/365099854_S_2_-Labeling_Shot-By-Shot_Microscopic_Badminton_Singles_Tactical_Dataset"}]

Based on the search results, I found comprehensive information about the ShuttleSet badminton dataset:

ShuttleSet is the largest publicly-available badminton singles dataset with annotated stroke-level records. It contains 104 sets, 3,685 rallies, and 36,492 strokes in 44 matches between 2018 and 2021 with 27 top-ranking men's singles and women's singles players.

ShuttleSet is manually annotated with a computer-aided labeling tool to increase the labeling efficiency and effectiveness of selecting the shot type with a choice of 18 distinct classes, the corresponding hitting locations, and the locations of both players at each stroke.

The dataset was presented in a research paper and appears in the Proceedings of the 29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD '23), August 6-10, 2023, Long Beach, CA, USA. The dataset and corresponding benchmarks are available at <https://github.com/wywyWang/CoachAI-Projects>.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

6. assistant (2026-06-06T17:34:07.010Z)

The claim is fully corroborated by the primary source (arXiv abstract) and multiple independent sources. Let me verify the exact verbatim wording matches.

The WebFetch of the arXiv abstract confirms every number, and the WebSearch independently surfaces the exact same statistics from the abstract, DeepAI, ResearchGate, and the KDD '23 ACM proceedings.

Verification against each element of the claim:

- "104 sets, 3,685 rallies, 36,492 strokes, 44 matches (2018-2021)" ? verbatim match to the abstract: "104 sets, 3,685 rallies, and 36,492 strokes in 44 matches between 2018 and 2021."
- "human-annotated stroke-level badminton singles dataset" ? matches the paper title: "ShuttleSet: A Human-Annotated Stroke-Level Singles Dataset for Badminton Tactical Analysis."
- "18 shot-type classes, hitting locations, and both players' locations at each stroke" ?

matches: "shot type with a choice of 18 distinct classes, the corresponding hitting locations, and the locations of both players at each stroke."

The claim is well-supported, current (KDD '23, peer-reviewed), and from a primary source. No contradicting evidence exists. The claim does not overreach beyond what the abstract states. (Minor note: the abstract specifies "27 top-ranking men's and women's singles players," which the claim omits but does not contradict ? this is an omission, not an error, and does not weaken any asserted fact.)

7. user (2026-06-06T17:34:12.268Z)

Structured output provided successfully

8. assistant (2026-06-06T17:34:14.872Z)

Verification complete. The claim is NOT refuted (refuted=false) with high confidence ? every figure matches the primary arXiv abstract verbatim and is independently corroborated by the peer-reviewed KDD '23 publication and multiple other sources.